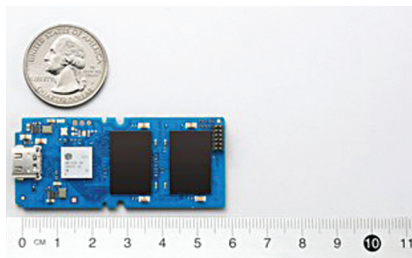


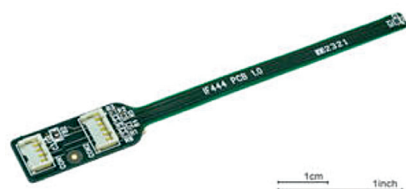
console users requiring high performance and the low power requirements of laptop users. The device features a USB 3.2 Gen 2 interface, advanced SSD controller architecture, and four NAND



channels that deliver sequential read/write speeds up to 2,100/2,000MB/s. By eliminating a USB bridge chip typically present in traditional designs, the SM2320 solution also lowers the Bill of Material cost and power consumption.
Silicon Motion Technology
www.siliconmotion.com

Frozen screen detector

Display Technology presents a new cost-effective sensor called IF444 that can detect a frozen screen. It can be integrated with a PC, SBC systems, or used in conjunction with a TFT controller such as Distecs Prisma cards. The motion sensor checks a few pixels of the active screen area for a blinking pattern. If the image signal is missing,



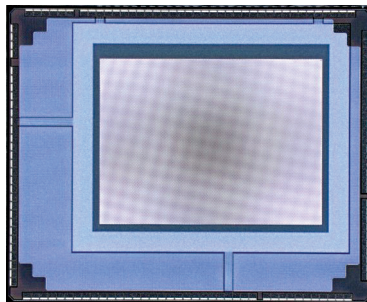
or the background lighting is defective, a warning signal is passed on to the PC or SBC system by Distecs Prisma and Artista TFT controller or the connectivity board IF445. An audible option is also available for this. The IF444 can be used both under Windows and Linux.

Display Technology
www.displaytechnology.co.uk

iTOF sensor for 3D imaging

Gpixel, in collaboration with Tower Semiconductor—a foundry of high value analogue semiconductor solutions—has launched GTOF0503 indirect Time of

Flight (iToF) sensor for 3D imaging that utilises Tower's pixel on its 65nm leading pixel-level stacked BSI CIS technology. The GTOF0503 features a 5µm, 3-tap state-of-the-art iToF pixel incorporating a



pixel array with a resolution of 640 × 480 pixels, offering superb performance for a broad range of fast-growing depth sensing and distance measurement applications such as vision guided robotics, bin picking, automated guided vehicles, automotive, and factory automation.

Gpixel
www.gpixel.com

Laser beam scanning sensor

MicroVision, a developer of MEMS based laser beam scanning technology, continues to build automotive lidar products on proprietary micro-electromechanical system solid-state technology. Based on time-of-flight architecture, the lidar



sensors are reliable, cost-effective, and scalable for automotive use. With a 250-metre detection range, ample reaction time gets delivered at highway speed. High resolution to identify objects at 20 million points per second at 30Hz, dynamic field of view (FOV), and velocity field output at 120Hz allows accurate prediction of uncertain driving situations even at high vehicle speeds.

MicroVision
www.microvision.com

GaN power transistors

GaN Systems, a provider of GaN power semiconductors, has introduced two

new GaN power transistors that come in an 8mm × 8mm PDFN package. The GS-065-011-2-L reduces the cost per watt of power delivered in 45W to 150W applications while the GS-065-030-2-L offers



advantages of low-cost GaN in applications up to the 3,000W power level. These new parts add to the GaN Systems family of low-cost GaN transistors that help designers improve efficiency, thermal management, and power density with increased design flexibility and cost effectiveness to meet new demands from consumer, industrial, and data centre customers.

GaN Systems
www.gansystems.com

High-performance AI chipset

Designers of high-performance, high-power artificial intelligence (AI) systems can now efficiently reduce power, cost, heat, and total solution size with the MAX16602 AI cores dual-output volt-



age regulator and the MAX20790 smart power-stage IC from Maxim Integrated. Leveraging the proprietary current ripple cancellation coupled inductor, this AI multi-phase chipset improves efficiency and enables greater than 95% efficiency at 1.8V output voltage and 200A load conditions. This increase in efficiency translates to a 16% reduction in wasted power. AI systems implemented with the MAX16602 and MAX20790 multi-phase chipset generate less heat and enable AI computing at the edge.

Maxim Integrated
www.maximintegrated.com